

Monotone softplus-PWL maps

slope = softplus(raw) $\times$ tanh(raw_sign) — positivity of magnitude, learned direction

Shared hinge slope constraint

$$m = \text{softplus}(s) \cdot \tanh(\sigma) \Rightarrow |m| = \text{softplus}(s) \geq 0, \text{ sign}(m) = \text{sign}(\tanh(\sigma))$$

Trend

$g(t) = b + \sum_k m_k \cdot \text{ReLU}(t - c_k)$
shared sign σ across hinges
no Level-1 attention

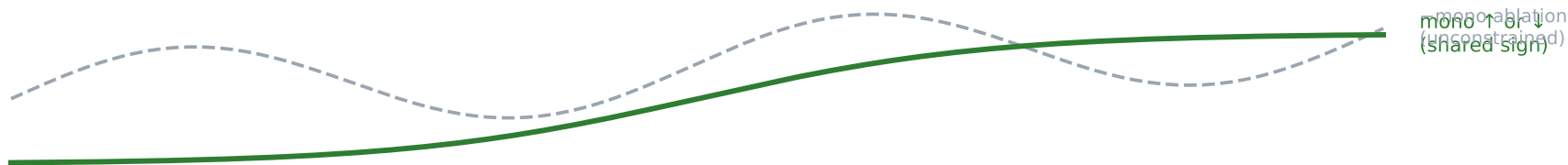
Holiday

per holiday h : $|d_h| \rightarrow \text{PWL mono}$
 $m_h = \text{softplus}(s_h) \cdot \tanh(\sigma_h)$
then Level-1 over holidays

Regressor

per lag/state channel j :
 $x_j \rightarrow \text{PWL mono in } x_j$
 $m_j = \text{softplus}(s_j) \cdot \tanh(\sigma_j)$
then Level-1 over channels

Effect on response shape (fixed SKU)



SKU FiLM uses softplus scale so per-SKU affine personalization preserves monotonicity in the structured input.